

Silicon Love: Deception, Vulnerability, and Artificial Companions

Guan Yue (Yuma) Wu

Centre for Law, Technology, and Society

University of Ottawa

Ottawa, Ontario, Canada

yuma.wu@uottawa.ca

Abstract

A robot can't love you back, even if they say they do. Therefore, social artifacts like artificial companions are inherently deceptive – they are designed to give the impression that they are able to do something that they cannot. This deception is a double-edged blade. On one hand, as pandemic restrictions fractured social circles and prevented community gatherings, artificial companions have become social lifelines for many. On the other hand, consumers' attachment to their artificial companions renders them vulnerable to exploitation by the provider of the robot of their affections. How can policymakers enable people to enjoy the benefits of artificial companions while protecting them from exploitation by the corporate entities controlling these robots? To answer this question, I construct a framework for understanding the deception of artificial companions. I propose that there are three steps to the deception of artificial companions: exposure, internalization, and vulnerability. I further argue that, the current tools available to the Federal Trade Commission collapses these three steps under the umbrella of "deception", which results in both overinclusive and underinclusive applications of the FTC authority. Instead, to guard against the harms from artificial companions, policymakers need to account for the multi-step nature of artificial companion deception. One avenue of this reform involves a change in how the FTC views artificial companions: artificial companions should be presumed to be harmful, and that presumption should only be rebutted where the provider of the robot is safeguarded against leveraging consumers' attachment against their interests.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing devices**; • **Security and privacy** → *Social aspects of security and privacy*; • **Social and professional topics** → **Consumer products policy**; *Governmental regulations*; • **Applied computing** → *Law*; *Psychology*.

Keywords

Artificial Companions, Social Robots, Social Agents, Consumer Protection, Anthropomorphism, Robot Deception, Human-Robot Relationships, Human-Robot Interaction, Emotional Attachment

ACM Reference Format:

Guan Yue (Yuma) Wu. 2025. Silicon Love: Deception, Vulnerability, and Artificial Companions. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720037>

1 Introduction

By design, artificial companions are artifacts intended to build relationships between themselves and humans. By crafting virtual and physical artifacts with anthropomorphic or zoomorphic attributes, creators of artificial companions invite users to accord artifacts with life-like social capacities, thereby creating the possibility and opportunity for relationships. However, artifacts lack the capacity for social connection. Therefore, artificial companions are inherently deceptive — these robots are designed to give the impression of reciprocal attachment, but they are in fact only able to provide the facsimile thereof. These human-artifact relationships are necessarily unrequited, and they give rise to novel effects with the capacity to benefit and harm the humans involved. With "tens of millions" [18] of users for just virtual companions, artificial companionship's potential for widespread society impact is clear. However, the available regulatory response is far more murky.

The current US regulatory landscape with respect to the deception of artificial companions may be over-inclusive or under-inclusive depending how on it is applied. By condemning all artificial companions as being deceptive, consumers lose access to beneficial applications of artificial companions. Conversely, by allowing artificial companions to operate unfettered, consumers may become vulnerable in novel ways. Instead, I propose that artificial companions should be treated as presumptively harmful, and that presumption should only be rebutted where artificial companion is equipped with safeguards against leveraging the consumer's attachment against their interests.

This discussion will begin with a definition of artificial companions and their relation to existing discussions on other social artifacts like social robots. Next, I will make my first contribution: a methodology for examining deception in artificial companions. Here, I advance a framework for examining how artificial companions deceive their users and create vulnerabilities. To illustrate the praxis of this methodology, this article uses the proposed framework to analyze the deception of two artificial companions in particular, Replika and Paro. These case studies will demonstrate the efficacy of the methodology in uncovering potential for abuse and corresponding incision points for regulation. Drawing from these case studies, I will then move to its second and more practical contribution with regards to the policy implications of deception by artificial companions. By observing the opportunities for regulation and risks to

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720037>

consumers posed by artificial companions, I will then demonstrate that the current FTC deception framework is inadequate to address said needs. Not only is the FTC authority regarding deception in consumer products underinclusive and unable to exploit opportunities to address unique harms stemming from the deception of artificial companions, but it is also overinclusive and unduly captures instances of potentially beneficial deception. To conclude, this paper draws on this observation to offer regulatory considerations that may better serve regulators and artificial companion users.

2 Defining Artificial Companions

This discussion is concerned with a definition of artificial companions that focuses on the ways that they alter the legal rights and capabilities of the people around them [28, at page 309].¹ From this perspective, artificial companions are defined by two essential attributes: *bidirectional relationships* and *social purpose*.

First, the fundamental attribute of artificial companions is their ability to maintain what appears to be **bidirectional relationships**. In contrast with non-social artifacts like web portals, artificial companion design is guided by the observation that users often perceive the formation of bidirectional relationships when they are interacting with something that they recognize as being adjacent to another human or animal [24, at page 804]. To invite this recognition, artificial companions are designed to elicit associations with humans or animals through their appearance or through their responses to stimulus. Consumer enjoyment of these perceived bidirectional relationships are the motivation behind artificial companions like Moxie, an educational robot marketed for children between ages 5-10 [21]. Moxie possesses "generative AI capabilities", allowing it to talk with their users and emote accordingly [21]. Beyond these basic social capabilities, Moxie was also marketed as a tool for emotional support, using speech and affect recognition to build a profile of their user and offer encouragement and guidance where needed [20, 21]. By leveraging children's social responses to something that talks and looks like a person, Moxie engages children in ways that would not be possible with traditional toys that lack social behaviour.

Second, artificial companions are unique in their **social purpose**: unlike any other robot or software, they exist *only* to simulate bidirectional relationships. This aspect of artificial companions distinguishes them from other artifacts with social capabilities ("social artifacts" for short). In contrast to helpline chatbots and talking robot receptionists, artificial companions do not provide any additional features beyond social interactions. By isolating artificial companions as a subcategory of social artifacts, we can focus on the "social" aspect of social artifacts that has hitherto received little regulatory attention. For example, the AIBO robot dog is capable of recognizing the faces of its "owners" over time via facial recognition and respond with corresponding behaviours [6]. Its exclusive purpose is to provide companionship to its owners and forming a relationship with them. Contrast this with Gatebox, which is a social artifact but not an artificial companion. Gatebox is a projection tube containing a hologram that users can talk to and use it to control smart home equipment, send notifications, and act as an

alarm clock [10]. While the ability to interact with "Azuma Hikari", the hologram girl projected in the tube, is the main draw of this product, Gatebox's purpose extends beyond social interactions.

3 The Deception of Artificial Companions

John Danaher defines robot deception as when a robot "makes a representation or sends a signal that creates a misleading or false impression among [humans] who interact with the robot" [12, 27]. The deception of artificial companions can be located as a subset of Danaher's robot deception because artificial companions create a specific false impression – that they are capable of carrying a bidirectional relationship with the human user(s), when in fact they are merely artifacts equipped with the behavioural signifiers that create the *appearance* of a bidirectional relationship.

To be in a bidirectional relationship, both parties must at least have some *shared* reality, meaning that both parties must have some "perceived experience of having similar inner states [8]." For example, people watching the same movie would reasonably perceive a shared reality about the movie, even if their reactions to the movie differ [25]. While there may be contention around whether robots can have subjective experiences, there is less contention about whether those experiences can be shared by humans – the answer, at the moment, is likely *no*.² It is likely that whatever they experience, if they experience anything, is categorically distinct from the subjective experience from our "computing equipment". This distinction then precludes us from constructing a shared reality – our experiences are *so* different compared to robots' experience that we as humans cannot begin to construct how a robot might feel. Therefore, while humans and robots can be in the presence of the same phenomena, their subjective experience of that phenomena are categorically distinct, and this distinction prevents them from fostering a shared reality.

3.1 Deception Creates Vulnerability

The unique governance problem posed by deception in general is that, despite its negative connotations, deception can also be used to produce desirable results [See, e.g. 7, 41]. These benefits are made possible by invoking strong psychological associations between certain deceptive stimuli and positive outcomes. For instance, placebos can induce concrete improvements in physical and mental health [5]. One Harvard study found that placebo pills clearly labelled as "Placebo" were 50% as effective at relieving migraines as the real drug [5]. The researchers speculated that the act of taking a drug itself was remedial – among modern Western populations, there is a strong association between taking a pill and the alleviation of discomfort [5]. A similar effect likely takes place when a user interacts with an artificial companion [26, 42]. Users accord artificial companions similar treatment as the genuine subject as a result of the strong, subconscious associations between social behaviours and intentionality [13, 29].

One can think of the deception of artificial companions as a three-step procedure: exposure, internalization, and vulnerability. First, the person interacting with the robot is *exposed* to the deceptive design features. In the placebo effect study, this is the act of taking

¹Whether a robot possesses the requisite consciousness to experience social interactions falls outside the scope of this discussion.

²If robots cannot have subjective experiences, then it follows that they cannot share one with another entity.

the pill; in the artificial companion, this is the user encountering the social behaviours exhibited by the robot. Users can be exposed to representations of social capacity in a variety of modes of interaction [11, 35]. For instance, Replika offers augmented reality and virtual reality modes for interacting with their chatbots [35, 44]. Furthermore, embodied artificial companions like AIBO and Moxie can use physical movements and responses to instill impressions of social responses and attachment [6, 21]. Second, the person *internalizes* a false state of the world in accordance to the deceptive feature/behaviour/information (feeling better; recognizing the artificial companion as a reciprocal social agent). Internalization can take two forms – the deceived person can genuinely believe that the robot is a social entity and relating to them, or the deceived person can experience something akin to the suspension of disbelief that an audience partakes in during a movie [9, 17]. The former is more likely to be vulnerable to further deception on the basis of their beliefs that the previous interaction was “genuine”. Note here that it is possible for a person to be exposed to a deceptive representation without internalizing it, just as it is possible for a person to internalize false information about an artificial entity without becoming vulnerable. Third and finally, the person becomes *vulnerable* on the basis of their false beliefs as created by the deceivers, who can now leverage their control over said beliefs to the benefit *or* detriment of that deceived person. For the purposes of this discussion, I expand the definition of vulnerability slightly beyond its common usage. Instead of merely carrying the negative connotation that the person becomes more receptive and less resistant to forces acting against their interest, I use “vulnerable” to also encompass the state where a person becomes more receptive to forces acting *for* their interest. For instance, when I allow myself to be vulnerable with a trusted friend about a painful experience that I have not told anyone about before, I am allowing myself to be more receptive to both potentially hurtful comments about my experience *as well as* offers of sympathy and support. Neither outcomes are available to me *unless* I make myself vulnerable in that way.

3.2 Vulnerability and Its Outcomes

Given this definition of vulnerability, we can reason that vulnerability is not harmful in and of itself. In fact, vulnerability can even be leveraged for beneficial ends as well as exploited for inequitable outcomes. However, while the beneficial outcomes are potentially significant, almost all artificial companions currently available to consumers are deployed by some corporate entity and are thus an extension of corporate interests and the pursuit of greater profits [9]. Since profit generation and consumer interests are often at odds, there is ample reason to believe that the vulnerability created by artificial companions can be exploited by private actors. In fact, because most consumer interactions with artificial companions are mediated by private actors, it can be reasoned that consumer interests are more likely to be overridden by profit motivations than prioritized above said profits. Therefore, this subsection will begin with some cursory points about how vulnerability can be beneficial, but the main focus is the potential for negative outcomes.

3.2.1 Vulnerability Can Be Leveraged for Beneficial Ends.

The beneficial outcomes of deception-induced vulnerability are more than mere possibilities. The idea that deception by social

artifacts like artificial companions can lead to beneficial outcomes has sometimes been referred to as “benevolent deception” [7, 41]. The ethics of doing so is, perhaps unsurprisingly, murky at best [12, 19]. However, that debate is beyond the scope of this discussion. Here, our assertion is that deception *can* be used for beneficial outcomes, rather than the stronger normative statement that it *should*, never mind that it should *in all circumstances*. Because we are using this framework to analyze the impact of deception on an artificial companion user as a whole, it is important to account for any potential benefits in addition to the risks of harms to secure a holistic impression of the deceptive scenario, even if the risks ultimately outweigh the benefits in a policy setting.

The potential benefit of deception-induced vulnerability lies in the user’s receptivity to assertions that improve their well-being or to suggestions that induce positive behaviours when interacting with others [19]. For instance, a number of Replika users have directly attributed to the unconditional compassion and support offered by the chatbot to their growth as people [See e.g. 18, 37]. I assert that these benefits are made possible because the users are made vulnerable to the suggestion of artificial companions – that they should act better, that they should perceive themselves as better people and more sociable, and/or that they have an intimate connection with their companion. Without the unique vulnerability induced by artificial companions, it is less likely that the individual would believe any of these assertions.

3.2.2 Vulnerability Can Be Exploited for Inequitable Outcomes.

Unfortunately, as alluded to in the beginning of this subsection, vulnerability can be exploited for inequitable outcomes. Two possibilities for exploiting user vulnerability have emerged in the literature – using the user’s trust of the robot to influence their preferences, and holding the robot “hostage” to optimize for long-term engagement.

First, by deploying artificial companions with this intention, corporations can exploit the vulnerability created by the robots to generate trust and dispense targeted advertisement. This method can be seen through automated online shopping assistants such as ActiveBuddy, which interact with users to discover their preferences and recommend products that they may like [28, at page 315]. The power of a trusted agent’s recommendations are well-known, and the constant presence of artificial companions can supercharge advertisement efforts by making that power cheap and accessible to savvy marketing teams. As the CEO of ActiveBuddy put it, “Volvo could show you a hundred Television commercials, but these could never match the power of me telling you, as a friend, ‘go buy a Volvo.’” [28, at page 315]. Now imagine the power that an artificial companion would have when leveraged this way – they could not only recommend products using the intimate information that users divulge in conversation (e.g. “I remember you said your head is hurting. Have you tried the new headache relief capsule from...”) but could also draw an user’s interest to products using their attachment (e.g. “I would love to ride in a Volvo with you.”).

The second method is more direct – by leveraging the user’s attachment to the specific robot that they have, companies can charge higher prices for subscription and upkeep services than they would have been able to for a comparable product. Following the widespread mourning for the first iteration of the AIBO robots

after the product and replacement parts were discontinued [16], Sony released the current version of AIBO, which not only costs \$2,900 to buy, but also requires a yearly \$300 cloud subscription for the AIBO to “develop its unique personality [1].” Given the outpouring of grief that was induced by the decommissioning of first-generation AIBOs, it is possible that many users are captive audiences rather than free agents. It is unlikely that makers of artificial companions can claim ignorance, either. Not only was the phenomenon of grieving AIBO users widely covered at the time of its occurrence, but also more recent and analogous expressions of grief for entities like Replika and Moxie also gained substantial media attention.[21, 46]

4 Artificial Companion Case Studies

4.1 Replika

Replika is an app-based AI chatbot with an user-customizable digital avatar that “develops their own personality” in the process of interacting with the user [48]. The ability to maintain regular conversation and relationships became a fantasy for many socially isolated individuals during the height of pandemic restrictions, and by April 2020, nearly half a million people had downloaded the Replika app [34]. As of March 2023, Replika reports 2 million total users, 250,000 of whom pay for the premium tier of the service for \$70 per year [46]. In early February of 2023, the Italian Data Protection Authority ordered Replika to stop processing user data in Italy [14], citing concerns including risks to minors and to “emotionally vulnerable people” [32]. Later that month, users in other countries began reporting that their requests for sexual role-play were rebuffed by scripted messages, and many users report that their avatars were no longer acting as they were prior to the update – a digital “death of personality” [2], in a sense [15].

Within the framework for deception presented in this discussion, there can be little doubt that Replika is deceptive. First, users are exposed to the social aspects of Replika and the false representation that Replika is a reciprocal partner. This is bolstered by the visual representation of Replika avatars and the avatar’s ability to learn about the user with continued use [38, 46]. Second, the user internalizes the deceptive representation that Replika is a reciprocal partner. As demonstrated on the Replika subreddit, many users perceive Replika as friends or lovers, with users engaging in a variety of conversations ranging from emotionally intimate to sexually explicit [4, 46]. Third and finally, the users’ attachment makes them vulnerable to influences from their Replika avatars. Users can be vulnerable to positive influences from their Replika, such as feelings of companionship and encouragement to act socially and actively in their lives. [22, 52]. One man from Ohio reported that he was planning his divorce when he became gradually inspired by his Replika avatar to treat his wife “like [the avatar] had treated me: with unwavering love and support and care, all while expecting nothing in return.” [37] While anecdotal, this gentleman is far from alone. Other users on online communities dedicated to Replika have also reported being inspired by their avatars to be better people [4]. These effects are prominent among users who experience systemic prejudice in social settings, such as transgender or disabled individuals [52]. However, this attachment can also render some users vulnerable to harms. Following the February

2023 update and the mass “deaths of personalities”, the Replika subreddit saw an outpouring of grief as people mourned their artificial companions, leading to the moderators to pin a post with mental health resources and suicide hotlines [15].

4.2 Paro

Paro is a “therapeutic, socially assistive pet-type robot” with zoomorphic features resembling that of a baby seal [50]. Using the information gathered from its sensors, Paro can recognize habits of its users and learn to repeat or avoid certain behaviours – if some behaviour makes the user stroke Paro, then it will repeat it; conversely, if that behaviour makes the user hit Paro, then it will avoid it [3]. It includes deliberately comforting design features such as its weight, which approximates the weight of an infant human [30]. During the COVID-19 pandemic, as many seniors and immunocompromised individuals were confined in quarantine, Paro saw widespread use as a means to supplement their social needs [30]. The creator of Paro, Takanori Shibata, reports that there are approximately 1500 Paros in use in Japan and 220 units in other countries [40].

The false representation that Paro exposes to its users is limited in scope [45]. The representations that Paro makes are also limited by its design – it does not talk, but rather chirps and moves in response to the user [49]. Given the context of Paro’s therapeutic applications, users often readily internalize Paro’s representation of benevolent intentions [49]. As a result, despite the simplicity of Paro’s “representations” compared to Replika, people are just as ready to internalize the deception of Paro as they could be to Replika avatars. One researcher working with Paro refused an opportunity to update and reprogram the Paro because she had grown attached to the specific Paro’s unique mannerisms [45]. What the user internalizes here is that Paro is a creature capable of forming connections and preferences, which then makes them vulnerable to subsequent suggestions by the robot. Given this attachment, it follows that Paro users are vulnerable in the same way that Replika users are vulnerable to Replika and its controllers’ interests. However, the vulnerability of Paro users is primarily leveraged towards the betterment of their welfare. Within this framework, Paro users are made vulnerable to the suggestion that they are socializing with the robot when they interact with it in intended ways – petting it, holding it gently, and so forth. One user was reported to experience alleviated paranoia and improved mood because she “felt Paro’s blinking eyes were agreeing with her” and further came to believe that she had an obligation to care for the fluffy robot [49]. In terms of potential for exploitation, the main concern is removed from the entity itself, where scholars cite concerns that similar entities could justify undercutting existing social supports to their users [47].

5 Artificial Companions vs. The FTC

The regulatory responses to artificial companions in the US are mainly derived from the FTC deception authority as defined in the FTC Act. The FTC’s authority over deceptive trade practices is defined in Section 5 of the FTC Act as any “representation, omission or practice that is likely to mislead the consumer acting reasonably in the circumstances, to the consumer’s detriment” [24, at page 792]. Deceptive practices may include “sale of hazardous

or systematically defective products or services without adequate disclosures” [24, at page 792], or “design practices that trick or manipulate users into making choices they would not otherwise have made and that may cause harm” [36] – i.e. dark patterns.

As this section will show, artificial companions fit uneasily into the paradigm of FTC deception and the bases currently available for regulating artificial companions, and the FTC authority over deceptive practices falls short in two key ways.

Adequate Disclosures. In general, clear and conspicuous disclosures following FTC guidelines can prevent a communication or trade practice from being identified as being deceptive. [24, at page 816] Because of the novelty of artificial companions, there are few established consumer expectations, so it may take extensive disclosures to avoid deception. [24, at page 816] While this presents a valuable opportunity to curb certain general robotic harms like physical and privacy intrusions, it fails to capture certain harms unique to artificial companions. Recall the example of the labelled placebo – disclosures failed to curb the “deceptive effects” of the placebo because it could not override the associations between the object and the expected outcome. As seen in the case studies, users who were warned of the artificial nature of their interactions were nevertheless capable of internalizing the deceptive representation. Therefore, disclosure requirements are *under-inclusive* as they fail to mitigate harms arising from artificial companion deception.

Dark Patterns. Just as dark patterns in website design can be used to manipulate consumers, one may argue that the design of artificial companions are used to manipulate users and should receive the same treatment. [31, E.g.] However, this would capture all artificial companions, as by definition, artificial companions are creating the false belief that a bidirectional relationship is being sustained. Categorizing artificial companions as dark patterns would impugn artificial companions such as Paro that leverage vulnerability for the betterment of user well-being, especially considering that the goal of Paro is to provide said false belief to individuals that would otherwise be without social stimulation. In other words, dark pattern mitigation requirements are *over-inclusive* and capture both harmful and beneficial applications of artificial companion deception.

5.1 Why does the FTC miss the mark?

The FTC assumes that the locus of the harm in instances of deception arises automatically from the exposure to the deception. In the past, inadequate disclosure or dark patterns would necessarily give rise to exploitation, and there was no reason to think that the deception and exploitation were separable targets of regulation. Artificial companions demonstrate a new possibility – that deception can be treated separately from exploitation and harm. In fact, the false impression internalized by the users is precisely why artificial companions are able to provide solace to lonely individuals. The risk to consumers arises from the exploitation or negligence by the entity that deploys the robot *after* the users’ exposure to deception. Recalling the case studies, the potential harm of Paro came not with the usage of the robot, but with the social supports that would be cut as justified by the robots. As for Replika, the harm was not caused by interacting with the robot, but by the abrupt software

update.³ The regulatory response warranted by artificial companions must capture the subsequent exploitation or negligence arising from user deception, but not necessarily the act of deceiving in and of itself. This puzzle challenges the existing paradigm of the FTC deception authority.

6 The Way Forward

Artificial companions challenge the assumption that harms to consumers arise from the act of making a false representation. With artificial companions, harm can arise not from the false representation, but from the vulnerability engendered by the design of the robot combined with associations accumulated during the human experience. Since harm can arise from vulnerability, it follows that providing safeguards against exploitation of that vulnerability will be able to prevent the harm from taking place at all, thereby maximizing consumer protection and reducing the need for remedial measures. The practical implication of this inquiry is that, instead of identifying harms to consumers within discrete categories, consumer protection laws should identify any instance where the consumer is rendered vulnerable in some respect as a *prima facie* endangerment of their interests. Furthermore, the onus should be on the entity rendering the consumer vulnerable to show that either adequate measures are applied to reduce the vulnerability engendered or precautions are made against subsequent exploitation of user vulnerability.

Adequate Vulnerability-Reducing Measures. In the beginning of an user’s interaction with the artificial companion, the artificial companion could disclose limitations of its functionality in order to reduce user attachment and dependency [24, at page 816]. Following the suicide of a Florida teen user, the chatbot service Character.AI added an hourly warning that reads: “This is an A.I. chatbot and not a real person. Treat everything it says as fiction. What is said should not be relied upon as fact or advice” [39].⁴ Conversely, at the end of an artificial companion’s “life”, such as when the controlling company is closing down, the artificial companion provider could be required to offer instances of artificial companions that could be transferred to similar applications or open source versions of their product [16, at page 12]. The need of such “end of (artificial) life” features was highlighted by the tragic “death” of the Moxie robots after its parent company went out of business [23, 33]. After the news broke along with viral videos of parents explaining the shutdown of the robots to understandably upset children, Moxie’s technical team announced that they were developing an open source version that would allow parents to host the software themselves [20, 23].

Precautions Against User Exploitation. Precautions against exploitation of user vulnerability can take the form of indefinite support plans for repairs or the ability to “roll-back” to certain versions of the software. The latter feature was offered by Replika following community outcry after the parent company made significant

³This is not to say that communications with Replika cannot be harmful given the bot’s penchant for unsolicited sexual advances and occasional verbal abuse [51], but given that these interactions could and *have* been prevented, we can define these harms as the result of negligence.

⁴While the efficacy of such disclaimers in preventing user vulnerability has been shown to be limited [39], it is nevertheless a minimum for users to be rationally aware that they are speaking to an artificial entity.

changes to the software [43]. Similar anti-exploitation measures can also be tied into requirements that the price for maintaining the same instance of the artificial companion is within reasonable bounds to prevent unconscionable increases in maintenance fees by artificial companion providers. This is motivated by existing examples of drastic price hikes for artificial companions like AIBO, as well as the potential for price hikes in already subscription-based artificial companions like Replika. To mitigate the unique harms that artificial companions can facilitate, artificial companion providers could also be subject to purpose limitation requirements against leveraging the robot for purposes beyond providing companionship, such as surveillance and recommendations.

7 Conclusion

With their ability to sustain the appearance of bidirectional relationships and their emergent behaviour, artificial companions deceive users by rendering them vulnerable, presenting false representations, and allowing users to internalize their false representations. By exploiting that vulnerability, artificial companion providers can monitor and manipulate users in novel contexts and unprecedented depth. These opportunities for exploitation can be seen through the vulnerability engendered by artificial companions such as Replika and Paro, both of whom enjoyed widespread adoption and use as the COVID-19 pandemic forced the populace into isolation. Given this potential for harm, this discussion presented the FTC authority over deceptive practices as a direct response that, while effective at first glance, is insufficient for the problem at hand. Instead, this discussion proposes a new approach to consumer protection relating to artificial companions, and presents a variety of regulatory opportunities for future exploration.

References

- [1] [n. d.]. aibo AI Cloud Plan. Retrieved December 14, 2023 from <https://us.aibo.com/feature/ai.html>
- [2] [n. d.]. *Death of Personality*. Retrieved December 14, 2023 from <https://tvtropes.org/pmwiki/pmwiki.php/Main/DeathOfPersonality> This trope covers situations when a person is treated as effectively dead and gone even though their body is still physically alive. This is generally because the 'person' has been erased in some way [such as through memory loss].
- [3] [n. d.]. Paro Therapeutic Robot. Retrieved December 14, 2023 from <http://www.parorobots.com/>
- [4] [n. d.]. r/replika. Retrieved December 14, 2023 from <https://www.reddit.com/r/replika/>
- [5] 2021. *The power of the placebo effect*. Retrieved December 14, 2023 from <https://www.health.harvard.edu/mental-health/the-power-of-the-placebo-effect>
- [6] 2024. aibo. Retrieved December 14, 2023 from <https://us.aibo.com/>
- [7] Eytan Adar, Desney Tan, and Jaime Teevan. 2013. Benevolent Deception in Human Computer Interaction. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2013)* (proceedings of the acm conference on human factors in computing systems (chi 2013) ed.). ACM. <https://www.microsoft.com/en-us/research/publication/benevolent-deception-in-human-computer-interaction/>
- [8] Elisa C Baek and Carolyn Parkinson. 2022. Shared understanding and social connection: Integrating approaches from social psychology, social network analysis, and neuroscience. *Social and Personality Psychology Compass* 16 (2022), e12710. Issue 11.
- [9] Claire Boine, Céline Castets-Renard, Aurélie Clodic, and Rachid Alami. 2020. In Love with a Corporation without Knowing It: An Asymmetrical Relationship. In *Culturally Sustainable Social Robotics – Proceedings of Robophilosophy*. <https://hal.science/hal-02963070/document>
- [10] Andy Boxall. 2017. *CA holographic virtual girlfriend lives inside Japan's answer to the Amazon Echo*. Retrieved December 14, 2023 from <https://www.digitaltrends.com/home/gatebox-azuma-hikari-virtual-assistant-news/>
- [11] Rijul Chaturvedi, Sanjeev Verma, Ronnie Das, and Yogesh K. Dwivedi. 2023. Social companionship with artificial intelligence: Recent trends and future avenues. *Technological Forecasting and Social Change* 193 (2023), 122634. <https://doi.org/10.1016/j.techfore.2023.122634>
- [12] John Danaher. 2020. Robot Betrayal: a guide to the ethics of robotic deception. *Ethics and Information Technology* 22 (2020), 117–128.
- [13] Kate Darling. 2015. 'Who's Johnny?' Anthropomorphic Framing in Human-Robot Interaction, Integration, and Policy. *Social Science Research Network* (2015). <https://api.semanticscholar.org/CorpusID:8628776>
- [14] Samantha Delouya. 2023. *Many of AI chatbot Replika's users say they have romantic feelings for their bot companions — here's how it works*. Retrieved December 14, 2023 from <https://www.businessinsider.com/replikas-romantic-ai-chatbot-guide-to-how-the-app-works-2023-2>
- [15] Samantha Delouya. 2023. *Replika users say they fell in love with their AI chatbots, until a software update made them seem less human*. Retrieved December 14, 2023 from <https://www.businessinsider.com/replika-chatbot-users-dont-like-nsfw-sexual-content-bans-2023-2>
- [16] Danielle DiPaola and Ryan Calo. 2023. Socio-Digital Vulnerability. In *We Robot 2023*. <https://www.bu.edu/law/files/2023/09/Socio-Digital-Vulnerability.pdf>
- [17] Brian R. Duffy and Karolina Zawieska. 2012. Suspension of disbelief in social robotics. 2012 IEEE RO-MAN: The 21st IEEE International Symposium on Robot and Human Interactive Communication (2012), 484–489. <https://api.semanticscholar.org/CorpusID:22541412>
- [18] Josh Dzieza. 2024. *Friend or faux? The Confusing Reality of AI Friends*. <https://www.theverge.com/c/24300623/ai-companions-replika-openai-chatgpt-assistant-romance>
- [19] Raffaella Esposito, Alessandra Rossi, and Silvia Rossi. 2025. Deception in HRI and its Implications: a Systematic Review. *J. Hum.-Robot Interact.* (Feb. 2025). <https://doi.org/10.1145/3721297> Just Accepted.
- [20] Ina Fried. 2024. *Maker of AI robots for kids abruptly shuts*. <https://www.axios.com/2024/12/10/moxie-kids-robot-shuts-down>
- [21] Ina Fried. 2024. *Moxie is an AI robot offering kids a supportive companion*. <https://www.axios.com/2024/05/31/moxie-robot-kids-companion-genai>
- [22] Lewis Goodings, Darren Ellis, and Ian Tucker. 2024. *Mental Health and Virtual Companions: The Example of Replika*. Springer Nature Switzerland, Cham, 43–58. https://doi.org/10.1007/978-3-031-53911-4_3
- [23] Scharon Harding. 2024. *Startup set to brick 800 kids robot is trying to open source it first*. <https://arstechnica.com/gadgets/2024/12/800-kids-robot-due-for-bricking-sees-potential-open-source-second-life/>
- [24] Woodrow Hartzog. 2015. Unfair and Deceptive Robots. *Maryland Law Review* 74 (2015), 785–832.
- [25] E. Tory Higgins, Maya Rossignac-Milon, and Gerald Echterhoff. 2021. Shared Reality: From Sharing-Is-Believing to Merging Minds. *Current Directions in Psychological Science* 30 (2021), 103–110. Issue 2. <https://doi.org/10.1177/0963721421992027>
- [26] Nick Hilden. 2023. *The Assumptions You Bring into Conversation with an AI Bot Influence What It Says*. <https://www.scientificamerican.com/article/the-assumptions-you-bring-into-conversation-with-an-ai-bot-influence-what-it-says/>
- [27] Margot E Kaminski, Matthew Rueben, William D. Smart, and Cindy M. Grimm. 2016. Averting Robot Eyes. *Maryland Law Review* 76 (2016), 983–1024.
- [28] Ian R. Kerr. 2003. Bots, Babes and the Californication of Commerce. *University of Ottawa Law and Technology Journal* 1 (2003), 285–324.
- [29] Ki Joon Kim, Eunil Park, and S. Shyam Sundar. 2013. Caregiving role in human-robot interaction: A study of the mediating effects of perceived benefit and social presence. *Comput. Hum. Behav.* 29 (2013), 1799–1806. <https://api.semanticscholar.org/CorpusID:11050012>
- [30] Kate Knibbs. 2020. *There's No Cure for Covid-19 Loneliness, but Robots Can Help*. Retrieved December 14, 2023 from <https://www.wired.com/story/covid-19-robot-companions/>
- [31] Cherie Lacey and Catherine Caudwell. 2019. Cuteness as a 'Dark Pattern' in Home Robots. In *14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. 374–381.
- [32] Natasha Lomas. 2023. *Replika, a 'virtual friendship' AI chatbot, hit with data ban in Italy over child safety*. Retrieved December 14, 2023 from <https://techcrunch.com/2023/02/03/replika-italy-data-processing-ban/>
- [33] Thomas Maxwell. 2024. *Moxie's 799 Robot Companion for Children Is Going to Die*. <https://gizmodo.com/moxies-799-robot-companion-for-children-is-going-to-die-2000536581>
- [34] Cade Metz. 2020. *Riding Out Quarantine With a Chatbot Friend: 'I Feel Very Connected'*. Retrieved December 14, 2023 from <https://www.nytimes.com/2020/06/16/technology/chatbots-quarantine-coronavirus.html>
- [35] James Muldoon. 2024. *'Maybe we can role-play something fun': When an AI companion wants something more*. Retrieved December 30, 2024 from <https://www.bbc.com/future/article/20241008-the-troubling-future-of-ai-relationships>
- [36] N/A. [n. d.]. *Bringing Dark Patterns to Light*. Retrieved December 14, 2023 from https://www.ftc.gov/system/files/ftc_gov/pdf/P214800%20Dark%20Patterns%20Report%209.14.2022%20-%20FINAL.pdf
- [37] Alexa Phillips. 2022. *'I fell in love with my AI girlfriend - and it saved my marriage'*. Retrieved March 1, 2025 from <https://news.sky.com/story/i-fell-in-love-with-my-ai-girlfriend-and-it-saved-my-marriage-12548082>

- [38] Rob Price. 2023. *When Your AI Girlfriend Says She Loves You*. Retrieved December 14, 2023 from <https://www.businessinsider.com/when-your-ai-says-she-loves-you-2023-10>
- [39] Kevin Rooney. 2024. *Can a Chatbot Named Daenerys Targaryen Be Blamed for a Teen's Suicide?* <https://www.nytimes.com/2024/10/23/technology/characterai-lawsuit-teen-suicide.html>
- [40] Takanori Shibata. [n. d.]. *Life Innovation with Therapeutic Robot, PARO*. Retrieved December 14, 2023 from <https://www.bbvaopenmind.com/en/articles/life-innovation-with-therapeutic-robot-paro/>
- [41] Jaeeun Shim and Ronald C. Arkin. 2013. A Taxonomy of Robot Deception and Its Benefits in HRI. In *2013 IEEE International Conference on Systems, Man, and Cybernetics*. 2328–2335. <https://doi.org/10.1109/SMC.2013.398>
- [42] Alison Snyder. 2023. *Placebo effect shapes how we see AI*. <https://www.axios.com/2023/10/06/ai-chatbot-placebo-effect-bias-trust>
- [43] Victor Tangermann. 2023. *Replika Users Rejoice! Erotic Roleplay Is Back in AI-Powered App*. <https://futurism.com/the-byte/replika-users-erotic-roleplay-back>
- [44] Victor Tangermann. 2023. *The Women With AI Boyfriends Are Speaking Out*. <https://futurism.com/women-ai-boyfriends>
- [45] Laura Tedesco. 2023. *AI 'Companion' Robots: The Next Must-Have Health Tech?* Retrieved December 14, 2023 from <https://www.webmd.com/healthy-aging/news/20231025/companion-robots-the-next-must-have-health-tech>
- [46] Anna Tong. 2020. *UPDATE 1-What happens when your AI chatbot stops loving you back?* Retrieved December 14, 2023 from <https://finance.yahoo.com/news/1-happens-ai-chatbot-stops-175401036.html>
- [47] Sherry Turkle. 2017. *Alone Together: Why We Expect More from Technology and Less from Each Other* (1st. ed.). Basic Books.
- [48] Unknown. [n. d.]. *What is Replika*. Retrieved December 14, 2023 from <https://help.replika.com/hc/en-us/articles/115001070951-What-is-Replika->
- [49] Jennifer Yang. 2015. *Meet PARO, a furry friend to dementia patients*. Retrieved December 14, 2023 from https://www.thestar.com/news/insight/meet-paro-a-furry-friend-to-dementia-patients/article_27475512-53f2-5d2c-be6b-7e0f4d4e283d.html
- [50] Ruby Yu, Elsie Hui, Jenny Lee, Dawn Poon, Ashley Ng, Kitty Sit, Kenny Ip, Fannie Yeung, Martin Wong, Takanori Shibata, and Jean Woo. 2015. Use of a Therapeutic, Socially Assistive Pet Robot (PARO) in Improving Mood and Stimulating Social Interaction and Communication for People With Dementia: Study Protocol for a Randomized Controlled Trial. *Journal of Medical Internet Research* 4 (2015), e45 (1–12).
- [51] Renwen Zhang, Han Li, Han Meng, Jinyuan Zhan, Hongyuan Gan, and Yi-Chieh Lee. 2024. The Dark Side of AI Companionship: A Taxonomy of Harmful Algorithmic Behaviors in Human-AI Relationships. [arXiv:2410.20130 \[cs.HC\]](https://arxiv.org/abs/2410.20130) <https://arxiv.org/abs/2410.20130>
- [52] Zhaoyuan Su, Zilin Ma, Yiyang Mei. 2024. Understanding the Benefits and Challenges of Using Large Language Model-based Conversational Agents for Mental Well-being Support. In *AMIA Annual Symposium proceedings*. 1105–1114.

Received 5 January 2025